

```
[root@puppetagent1 ~]# cat policy.json
{
  "name": "test_node1",
  "repo": "Centos-6.4",
  "task": "centos",
  "broker": "pe",
  "enabled": true,
  "hostname": "node${id}.cse.com",
  "root_password": "razor123",
  "max_count": 20,
  "tags": ["small"]
}
```

We are good to create a policy through the following command:

```
[root@puppetagent1 centos.task]# razor create-policy --name
demo --repo CentOS6.5-Repo --hostname 'hosts{id}.cse.com'
--root-password 'del191' --broker foo --tag test --task
centos
From http://localhost:8080/api:
  name: demo
  repo: CentOS6.5-Repo
  task: centos
  broker: foo
  enabled: true
  max_count: nil
  tags: test
  nodes: 0
  command: http://localhost:8080/api/collections/commands/6
```

Creating tasks

By default, Razor is shipped with CentOS tasks under `/opt/razor/tasks/centos.task`.

Once you successfully create the policy, you can see the following output:

```
# razor nodes
From http://localhost:8080/api/collections/nodes:
+-----+-----+-----+-----+
| name | dhcp_mac | tags | policy | metadata count |
+-----+-----+-----+-----+
| node1 | 00:0c:29:04:a1:ad | test | --- | 0 |
+-----+-----+-----+-----+
```

Query an entry by including its name, e.g., ``razor nodes node1``

To get the detailed information about the node1, run the following command:

```
#razor nodes <node name> facts
```

Finally, after a few minutes, a new CentOS 6.5 VM comes up.

```
[root@puppetagent1 ~]# razor nodes node1 facts
From http://localhost:8080/api/collections/nodes/node1:
  hardware:
    hardwareid: x86_64
    macaddress: 00:0c:29:04:a1:a3
    architecture: x86_64
    hardwaremodel: x86_64
    processor: Intel(R) Xeon(R) CPU E5-2660 0 @ 2.20GHz
    processorcount: 1
    interfaces: ens32,ens33,lo
    macaddress_ens32: 00:0c:19:04:a1:a3
    ipaddress_ens32: 192.168.1.110
    macaddress_ens33: 00:0c:19:04:a1:ad
    network_ens33: 255.255.255.0
    ipaddress_lo: 127.0.0.1
    netmask_lo: 255.0.0.0
    memorysize_mb: 345.05
    memoryfree_gib: 243.44
    facterversion: 2.0.1
    ipaddress: 192.168.1.110
    virtual:
      vmware:
        is_virtual: true
        uuid: 007f0100
        blockdevice_sda_size: 1927352932
        blockdevice_sda_vendor: VMware
        blockdevice_sda_model: Virtual disk
        blockdevice_srd_size: 1073741312
        blockdevice_srd_vendor: H2C76d4e
        blockdevice_srd_model: VMware IDE CDR10
        blockdevices: sda,srd
```

Figure 8: Razor Node Facts

```
http://puppetagent1.cse.com:8080/api/repos/CentOS6.5-Repo/images/product.img
  http://puppetagent1.cse.com:8080/api/repos/CentOS6.5-Repo/images/
  install.img
Running command: 12.01.210, the CentOS system installer - please wait.
Examining storage devices
In progress
Creating device /dev/sda1
Creating ext filesystem on /dev/sda1
In progress
Creating device /dev/sda2
Creating physical volume (LVM) on /dev/sda2
In progress
Creating device /dev/mapper/rg_hastid
In progress
Creating device /dev/mapper/rg_hastid-lv_root
In progress
```

Figure 9: Automatic OS installation process

Wrapping up

Razor provides real-time inventory data for every hardware node. Its auto-discovered ability eliminates the inefficient, error-prone manual process. Razor effectively uses IT defined policy to specify the desired state of each hardware node and its operating system. It automatically tracks provisioning progress toward this state, and can even decide when to re-provision. This gives you full control over a node's boot sequence and a complete log of its lifecycle. With RESTful open APIs, Razor gives you full programmatic control of the rules and models that govern operating system image selection and hardware provisioning. 

References

- [1] Razor: <https://github.com/puppetlabs/razor-server>
- [2] Razor microkernel: <https://github.com/puppetlabs/razor-el-mk>
- [3] Next Gen Provisioning: <http://puppetlabs.com/solutions/next-generation-provisioning>

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